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Multimodal AI Teaching Copilot for Real-Time Visual Learning Generation

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***Abstract*—Lecture slides are prepared before class and cannot follow what a teacher actually explains, while live captions follow the speech but show only text. This paper presents the design and implementation of a teaching copilot that listens to a live lecture and builds a structured slide deck on the classroom projector while the teacher speaks. Speech is transcribed locally with a Whisper-based recognizer. A sentence-embedding concept tracker and a rule-based gate decide when a bounded unit of transcript is sent to a large language model (LLM), which returns a schema-validated description of topic, subtopic, discourse relation and content. A deterministic planner, not the LLM, decides whether to update the live slide, continue on a new slide or open a new topic, using confirmation and dwell rules that keep the display stable. A composer maps the content to designed layouts with typeset formulas and openly licensed images ranked with CLIP, and the teacher can pause, edit, navigate and replace images from a control view. In a continuous 17-minute chemistry lecture on a laptop with a 4 GB GPU, the system built 35 slides across six detected topics, with median speech-to-text and LLM latencies of 0.66 s and 1.30 s; 81 of 88 units were interpreted by the LLM and the rest by a deterministic fallback. Concept simulation and narrative (video) visualization are planned as further modes.**

***Index Terms*—Educational technology, speech recognition, large language models, automatic slide generation, real-time systems, multimodal learning, human-in-the-loop**

I. Introduction

Slides prepared before a lecture fix the order and depth of the material in advance, so examples, clarifications and answers given during the class do not appear on the screen. Research on multimedia learning shows that students learn better from words combined with well-organized pictures than from words alone [1], [2], provided that the visuals reduce rather than add extraneous load [3], [4]. A display that mirrors the spoken explanation in structured visual form is therefore attractive, but producing it by hand during a lecture is not practical.

Speech recognition [5] and large language models (LLMs) [6], [7] now make it possible to follow a lecture as it happens. Used naively, however, they give an unstable display: an LLM that re-generates slides from a growing transcript reorders, renames and rewrites content every few seconds, adds facts that were never taught [8], and exhausts free-tier quotas within minutes. Live captioning avoids these problems but shows only text, without the structure of the lesson [9].

This work asks how a live, continuity-aware slide display can be generated from a teacher's speech on modest hardware with free services, while the teacher stays in control. The contributions are: (1) an implemented end-to-end copilot that turns live lecture speech into projected slides during the lecture; (2) an architecture in which the LLM is a gated, bounded and validated reasoning service inside a deterministic event-driven core, so that slide lifecycle decisions are testable rules; (3) a planner and composer that separate *what was said* from *how it is shown*; (4) a teacher control view for live correction; and (5) an evaluation on a real 17-minute spoken lecture with component measurements, together with a clear account of what is not yet implemented.

II. Related Work

Visuals in teaching. Dual coding theory [2] and multimedia learning research [1] explain why verbal and visual representations together support understanding, while cognitive load theory [3], [4] and studies of lecture slides [10] warn that dense or irrelevant visuals hinder it. The copilot therefore condenses speech into short typed layouts and adds images only where they depict something concrete.

Speech recognition and captioning. Whisper [5] and runtimes such as faster-whisper [11] allow real-time recognition on consumer GPUs, with voice activity detection (VAD) [12] for segmentation. Verbatim classroom captions help access but are hard to follow at speaking rate [9]; here students see structured content.

Automatic slide generation. Prior work treats slide generation as an offline document-to-slides task: PPSGen [13] selected sentences from papers, D2S [14] summarized sections by query, and DOC2PPT [15] generated slides with text and figures. A lecture instead arrives incrementally, is spoken and noisy, and must stay stable while watched. Topic segmentation [16] and sentence embeddings [17], [18] give boundary signals, used here as one input.

LLMs in education. LLMs open new possibilities for explanation and material generation [6] but can produce statements unsupported by the input [8]. Schema-constrained output [19] reduces format errors, not factual drift. The copilot therefore asks the LLM only for a bounded, structured interpretation of what was just said, checks it against the transcript, and leaves layout and slide lifecycle to deterministic code. Table I contrasts the approach with existing workflows.

TABLE I
COMPARISON WITH EXISTING APPROACHES

Approach	Input	When produced	Lecture continuity	Teacher control	Visual structure
Prepared slides (conventional)	Teacher authoring	Before class	Fixed in advance	Full, but offline	As designed
Live captioning [5], [9]	Speech	Live	None (text stream)	Little	Plain text
Document-to-slides [13]–[15]	Written document	Offline	Document sections	Edit afterwards	Text + figures
Prompt-based LLM slide tools	Typed prompt	Offline, on request	None across a lecture	Edit afterwards	Template layouts
Proposed copilot	Live speech	During the lecture	Tracked topic / subtopic, planner rules	Live: pause, edit, navigate, images	Typed layouts, formulas, images

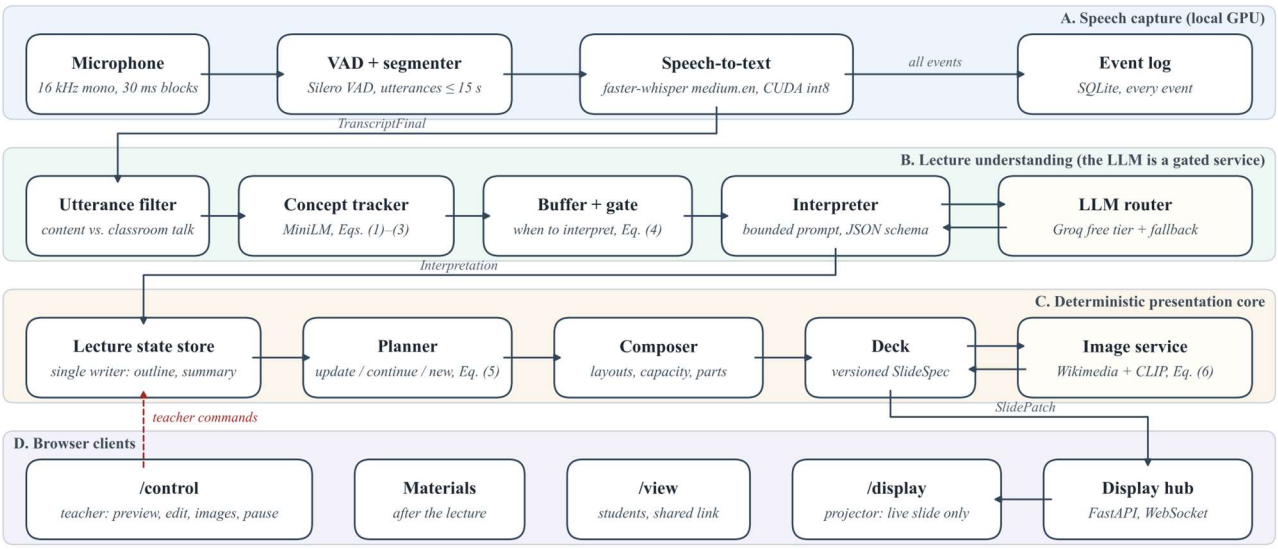


Fig. 1. Implemented architecture. Lanes A–C run in one Python process; the LLM is called only by the interpreter through the router. Browser clients receive versioned slide patches and send teacher commands back; the user interface never changes the core state directly.

III. SYSTEM ARCHITECTURE

Fig. 1 shows the implemented architecture: one Python *asyncio* process whose components communicate only through typed events on an in-process bus. A single store owns the lecture state, and every event is appended to an SQLite log, so each lecture can be replayed for debugging and post-lecture materials. Models run in worker threads with bounded queues, so audio capture never blocks.

The LLM is reached only through a gate, a rate limiter and a schema validator, and its answer is one input to rule-based decisions. Rendering is progressive (a quick deterministic update, then the refined interpretation in place, then images), and teacher commands override automatic decisions. Only free or local components are used: faster-whisper on the GPU, MiniLM and CLIP as ONNX models on the CPU, FastAPI with WebSocket, and browser clients with Preact, KaTeX and SVG. The LLM (gpt-oss-120b [7] with a Qwen backup) is hosted on the Groq free tier, because a 7B-class model does not fit beside the speech model in 4 GB of GPU memory.

IV. LECTURE UNDERSTANDING

A. Speech Capture and Concept Tracking

Audio is captured at 16 kHz in 30 ms blocks. Silero VAD segments speech (600 ms hangover, utterances capped at 15 s, segments under 300 ms dropped), faster-whisper transcribes each utterance with beam size 1 and a short subject-keyword prompt, and likely hallucinations are discarded. A deterministic filter then separates content from classroom management ("take out your books"), meta talk and fillers.

Each content utterance is embedded with MiniLM [18] (about 6 ms on CPU). With the L2-normalized embedding e_t and concept centroid c_{t-1}, the shift score and centroid update are

$$s_t = 1 - \mathbf{e}_t^{\top}\mathbf{c}_{t-1} \quad (1)$$

$$\mathbf{c}_t = \mathrm{norm}((1-\alpha_t)\mathbf{c}_{t-1} + \alpha_t\mathbf{e}_t) \quad (2)$$

where $\mathrm{norm}(x) = x/\|x\|$ and $\alpha = 0.35$, raised to 0.6 at a boundary so that the centroid re-seeds on the new concept. A boundary is signalled when

$$b_t = \kappa_t \vee (s_t \geq \tau), \tau = 0.75 \quad (3)$$

where κ_t marks a strong cue phrase ("now let us", "next", "is defined as"). On the fixture lectures the single-sentence shift was noisy (0.3–0.6 within one facet), so τ is high and cue words carry most boundary detection; the LLM decides the final relation.

B. Gate and Token Budget

Content accumulates in a buffer until the gate decides that a meaningful unit is ready. With p_t the VAD silence after a finished sentence, w_t the buffered words and a_t the age of the oldest pending line (times in seconds), the gate fires when

$$G_t = (p_t \geq 1.2 \wedge w_t \geq 6) \vee w_t \geq 40 \vee b_t \vee a_t \geq 12 \quad (4)$$

subject to an 8 s floor between calls (at most 7.5 per minute). New lines queue while a call is in flight, so interpretations never overlap; units end only after finished sentences, and fragments under six words join the next unit. The floor follows from the free-tier budget: with daily quota Q per key and model, system prompt P_s, dynamic context P_d and output O,

$$N_{\text{day}} \approx Q/(P_s + P_d + O) \approx 200000/2300 \approx 85 \quad (5)$$

or about 15–20 lecture minutes per key. Calling the LLM on every utterance would exhaust it several times faster.

C. Bounded and Truthful Interpretation

The interpreter never sends the whole transcript. The prompt holds a header (subject, level, an outline of at most 150 tokens), the current slide with numbered items, a rolling summary of at most 120 tokens and the new lines, checked against a 3,000-token budget. The LLM returns JSON with topic, subtopic, a relation (same concept, elaboration, sub-concept, sibling, new topic, digression), typed content items (definition, facts, classification, steps, comparison, timeline, formula, cause-effect, example), an image hint, revisions and *concerns* about possibly wrong statements. Pydantic validates the answer; after one failed repair, a deterministic fallback shows complete spoken sentences as points, and the fallback is logged.

Two guards keep slides truthful. A coverage guard compares every complete content sentence u (at least six words) with the model output O; with $W(\cdot)$ the set of word stems,

$$|W(u) \cap W(O)| \leq \tfrac{1}{3}|W(u)| \Rightarrow u \text{ is shown as said} \quad (6)$$

so a fact the model left out still reaches the slide. A grounding guard flags formula tokens that the model changed from what was heard. A correction is applied only above a confidence threshold (0.75 factual, 0.4 mis-hearing); otherwise the slide shows what the teacher said, and for factual concerns the teacher gets a card with both versions and a one-click switch. The router adds token buckets, a daily usage ledger, several free keys used in turn, and a 15 s deadline per unit.

V. Live Slide Generation

A. Planner

The planner turns each interpretation into one operation: *update* the working slide, *continue* on a new slide of the same topic (new facet or further part), open a *new* topic slide, *retitle*, or do nothing (Fig. 2). A new concept does not mean a new slide: an elaboration or sub-concept usually updates the live slide while it has room. A topic change must be confirmed, because one unit can be misread:

$$C_n = B_n \vee \left(\bar{s}_n^{\max} \geq \tau\right) \vee \left(T_n \equiv \tilde{T}_{n-1}\right) \quad (7)$$

where B_n is true if a boundary (3) occurred in unit n, $\bar{s}_n^{\max}$ is the largest shift score in the unit smoothed over two utterances, T_n the topic of unit n and $\tilde{T}_{n-1}$ an unconfirmed candidate from the previous unit. Unconfirmed content is shown tentatively and moved to the new topic's slide once confirmed. A minimum dwell of 15 s (6 s for the next part of a full slide) prevents rapid changes while pending slides keep merging content. Updates are additive and in place, near-duplicates (text similarity ≥ 0.85) are merged, and teacher edits are never overwritten.

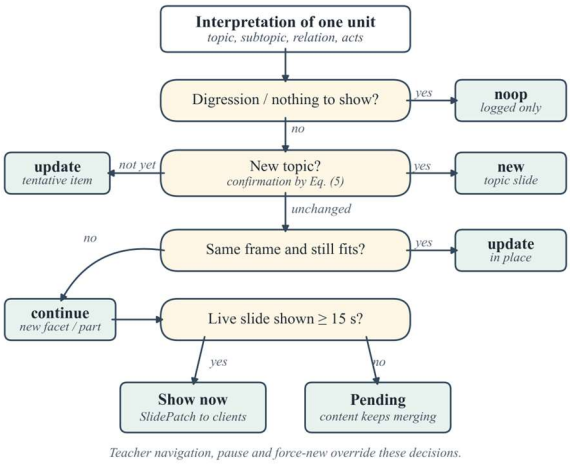


Fig. 2. Planner decision for one interpreted unit. The LLM supplies the relation and content; the decision itself is rule-based and unit-tested.

B. Composer and Layouts

The composer maps items to layout blocks: definition cards, key points, fact tiles, group and member cards, process flows (up to ten steps in two rows), comparison tables, timelines, cause-effect chains, classification trees and formulas. A pixel-level height model of the 16:9 stage decides what fits; overflow moves to a further part of the slide (I, II, …) instead of shrinking the text. Spoken mathematics is converted to LaTeX deterministically ("v equals under root 2RT upon m" becomes a square-root fraction) and typeset with KaTeX with a variable legend; chemical formulas get subscripts.

C. Image Retrieval

A policy decides whether a slide needs an image (concrete objects and structures do; abstract definitions, full-width diagrams and formula-led slides do not). For a query q suggested by the LLM, the service searches Wikipedia and Wikimedia Commons, filters by licence, size, aspect and English labels, and scores up to eight previews with CLIP [20] on the CPU. Image I is accepted when

$$\sigma(q, I) \geq 0.22 \wedge \sigma(q, I) - \max_{d \in D} \sigma(d, I) \geq 0.01 \quad (8)$$

where σ is the CLIP cosine similarity and D a set of generic distractors ("a logo", "a map"). Labelled diagrams rank first for structures, an image is placed only if the MiniLM similarity of query and slide text is at least 0.25, and the search has an 8 s budget; the teacher can find, change, upload or remove images at any time.

D. Live Display and Teacher Control

A FastAPI server pushes versioned slide patches over WebSocket to /display (projector, live slide only), /control (teacher) and /view (students on a shared link). In-slide updates animate only changed items, and a display reconnects

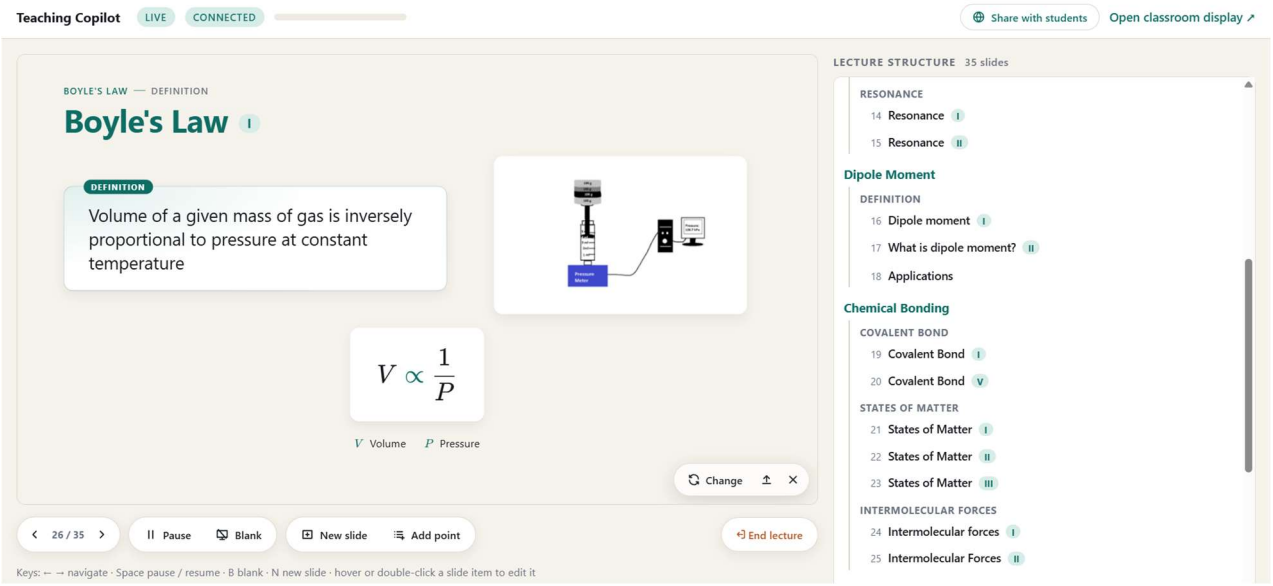


Fig. 3. Teacher control view during the live lecture of Section VI. Left: editable preview of the projected slide with an automatically retrieved apparatus image, the typeset formula with its legend, image tools and lecture controls. Right: the lecture structure tree built by the system (topic → facet → slides).

and requests the current state after a network drop. The control view (Fig. 3) offers an editable preview, navigation, pause (speech during a pause is not interpreted), blank screen, new slide, in-place editing, image tools, mistake cards, the lecture structure tree and the latest transcript line. Slides can switch between light and dark themes mid-lecture, and the control view has its own dark mode. On request after the lecture, the stored session yields summary and key-concept slides, notes and assignment PDFs, a PPTX copy and a searchable archive.

VI. Experimental Evaluation

A. Setup

All runs used a laptop with an NVIDIA RTX 3050 Laptop GPU (4 GB) and 16 GB RAM, and the free tier of Groq. The live lecture was spoken continuously by one of the authors into the laptop's microphone input with /control and /display open. It covered chemical bonding, then the gas laws and the kinetic theory of gases. The main model's daily quota had been spent in earlier tests, so the router served the session from the configured Qwen backup on the same provider. All numbers come from the application's log of this session; no user study was conducted.

B. Live Lecture Results

Table II summarizes the session. Over 17 min 26 s, 142 transcript segments formed 88 interpretation units. The planner made 55 in-place updates, 12 continuations and 7 new-topic decisions, and 14 units changed nothing (digressions, announcements, content already shown). Three topic proposals were held as tentative until confirmed by (7), the stability behaviour the planner is designed for. The final deck had 35 slides in the structure tree of Fig. 3.

TABLE II
Live Lecture Session (17 min 26 s, Chemistry)

Measure	Value
Transcript segments	142
Speech-to-text latency, median / p95 / max	0.66 / 0.95 / 1.82 s
Interpretation units (LLM / fallback)	88 (81 / 7)
LLM call latency, median / p95 / max	1.30 / 2.80 / 4.07 s
Relations: elaboration / same / sibling	46 / 20 / 10
Relations: new topic / sub-concept / digression	7 / 3 / 2
Planner: update / continue / new / noop	55 / 12 / 7 / 14
Tentative topics held for confirmation	3
Slides created (in final deck)	36 (35)
Mis-hearings corrected on the slide	23
Image searches / images found	25 / 8

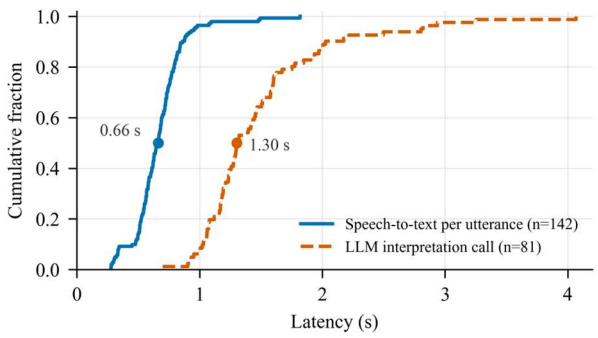


Fig. 4. Cumulative distribution of speech-to-text latency per utterance and LLM call latency per interpretation in the live session. Dots mark the medians.

Fig. 4 shows the latencies. Speech-to-text stays under one second for 95 % of segments; LLM calls have a median of 1.30 s. The delay seen in class is dominated by unit formation: in this session a line waited a median of 11 s before being sent, because every concept boundary produced a short unit that queued behind the 8 s floor. Units now end only after

finished sentences; replaying this lecture through the gate in simulation reduced the median wait from 9.5 s to 5.3 s, the worst case from 44 s to 18 s, and requests from 107 to 83 (verified in simulation and unit tests, not yet live).

Fig. 5 shows six of the 35 slides re-rendered from the recorded interpretations: a short process, term cards, a definition with its formula, and spoken formulas typeset with variable legends.

In Fig. 6, the seven fallback units (crosses) occurred when every key's daily quota was spent and the backup model's per-minute bucket was briefly full; the fallback still showed complete spoken sentences, and the router now waits for the first free bucket instead (unit-tested). Twenty-three mis-hearings were corrected silently, e.g. "collide with the world of container" shown as "walls of container" and "0 degree in 180" as "0 °C, 1 atm". An image was found for 8 of 25 searches; most misses were queries with no Commons file name matching every word ("pi bond orbital overlap"). A core-phrase search added later found images for 9 of the 16 failed queries, 7 clearly relevant on inspection.

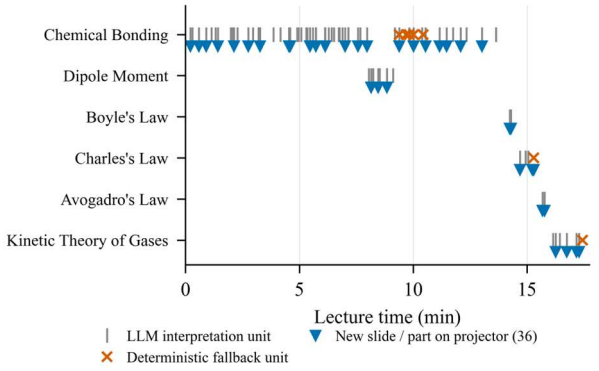


Fig. 6. Topics detected over the 17-minute lecture: LLM units (ticks), fallback units (crosses) and new slides or parts (triangles).

C. Component Measurements

On the same laptop, Whisper medium.en transcribes a 14 s clip in 0.75 s using about 1.1 GB of GPU memory; a MiniLM embedding takes about 6 ms and CLIP scores eight images in about 0.69 s on the CPU. On a 30-concept labelled set, the first returned image was relevant for all 28 concepts that returned one. On a scripted lecture with the main model, units were interpreted a median of 6.3 s (max 12.8 s) after their last segment, at most eight calls per minute and about 2.3k tokens per call. All 513 unit and integration, 28 browser and 8 GPU tests pass.

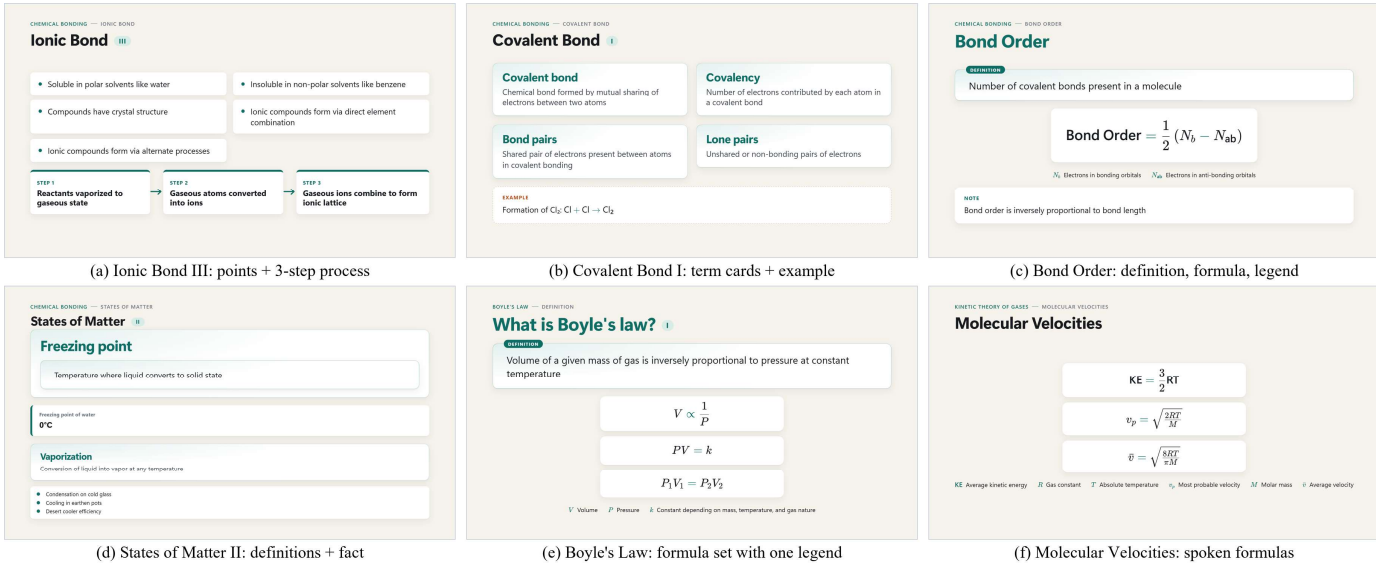
VII. Discussion

Unlike the approaches in Table I, the copilot combines live input, a stable deck that follows the teacher's sequence, and a human in the loop. Deterministic lifecycle decisions made the behaviour explainable: every slide change in the log carries a reason, and each issue found in live tests could be reproduced by replaying the recorded session at zero LLM cost. The bounded prompt and gate make a free-tier LLM usable, but only for limited lecture time per day; and latency is a pipeline property, decided more by unit formation and rate limits than by the models.

The live evaluation is a single 17-minute English lecture by one speaker with no classroom study, so no claim is made about learning outcomes. The session ran on the backup model, subtopic naming still varies between runs, image coverage depends on Wikimedia, and crash recovery and a 60-minute soak test are designed but not implemented.

VIII. Planned Visualization Modes

Because understanding is separate from rendering, further modes reuse the lecture state and add only their own planner, composer and display components. Two such modes are planned and not yet implemented. **Live concept**



(a) Ionic Bond III: points + 3-step process

(b) Covalent Bond I: term cards + example

(c) Bond Order: definition, formula, legend

(d) States of Matter II: definitions + fact

(e) Boyle's Law: formula set with one legend

(f) Molecular Velocities: spoken formulas

Fig. 5. Projector slides from the live lecture of Table II, re-rendered from the recorded LLM interpretations through the current display engine (no new LLM calls). (f) was built from the recognized lines "ke equals 3 by 2 times RT", "alpha equals under root 2RT upon m" (alpha corrected to v_p as a likely mis-hearing) and "average velocity v equals under root 8RT upon pi m".

visualization. For mechanisms with motion or forces, a slide block will host an interactive simulation in the spirit of PhET [21]. Developers write parametric templates (block on an incline, pulley, circuit, wave, orbit, pendulum, water and cell cycles); the LLM only selects a template and fills validated parameters such as mass, angle or the forces to show. SVG/Canvas rendering keeps it real time and prevents physics the template does not implement.

Narrative visualization. For historical or story-like content, the copilot will show a short visual narrative synchronized with the narration. Open text-to-video models [22], [23] need 8–24 GB of GPU memory and minutes per clip on a 4 GB GPU, and are not reliable for factual content. The first version is therefore a storyboard of three to six scenes from retrieved or teacher images with procedural camera motion and captions, first scene within 10 s; generated video is a later upgrade behind the same scene specification, prepared between lectures when enough GPU capacity is available. Quizzes through the shared link, the teacher's reference material, Hindi and Marathi input, and a classroom study are also planned.

IX. Conclusion

This paper presented an implemented teaching copilot that generates a structured, continuity-aware slide deck on the projector while the teacher speaks, treating the LLM as a bounded, validated interpreter inside a deterministic, event-driven core. On a laptop with a 4 GB GPU and free services, the system followed a continuous 17-minute chemistry lecture, built 35 slides across six topics and kept the teacher in control. Future work will measure its effect in real classes and add concept simulation and narrative visualization on the same core.

Acknowledgment

The authors thank the Department of Electronics and Telecommunication Engineering, JSPM's Rajarshi Shahu College of Engineering, Pune, for its guidance and support during this project.

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